

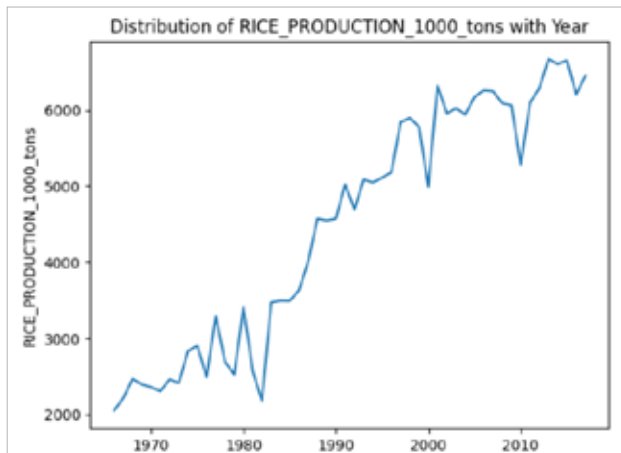


Figure 6: Distribution of rice production over the years

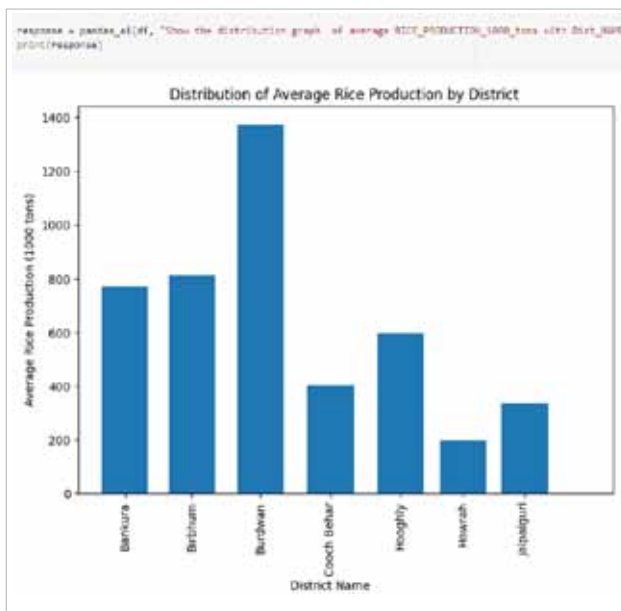


Figure 7: Distribution of district-wise rice production

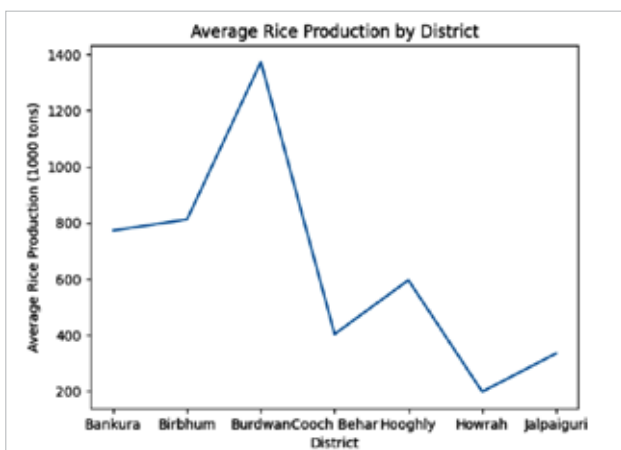


Figure 8: Line plot of average rice production by district

can easily be plotted with the *pandasai* module with the help of ChatGPT. For example, the distribution of rice production can be depicted as follows:

```
response = pandas_ai(df, "Show the distribution graph of RICE_PRODUCTION_1000_tons with Year")
print(response)
```

Similarly, the distribution of rice production in a district can be plotted with an instruction like:

```
response = pandas_ai(df, "Show the distribution graph of RICE_PRODUCTION_1000_tons with Year DIST_NAME Hooghly")
print(response)
```

A bar graph of the district-wise distribution of rice production can easily be drawn with an instruction like this:

```
response = pandas_ai(df, "Show the distribution graph of average RICE_PRODUCTION_1000_tons with Dist_NAME ")
print(response)
```

District-wise average rice production can be plotted like this:

```
df_sorted_rp=df.sort_values(by='RICE_PRODUCTION_1000_tons')
```

```
response = pandas_ai(df_sorted_rp, "line plot average RICE_PRODUCTION_1000_tons with DIST_NAME")
print(response)
```

PandasAI, a supplementary tool based on ChatGPT, enhances data analysis in the Pandas environment. Its unstructured query language support makes data set analysis easier. Due to this textual informal approach, it is a paradigm change in data exploration and subsequent analysis. Here, we have focused on a few examples of the application of PandasAI on a data set. Readers can explore the several possibilities of this textual query-based data analysis. Programs and the data set are available at https://drive.google.com/drive/folders/1gVvRcwDNDP4j_RbYdBm7Z0FKcUp95wNQ?usp=sharing. Interested readers can download the data and explore it further. **END** 🐼

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The author has more than 25 years of experience in software development. He has published two books on MATLAB and medical image processing (published by PHI Learning and SPD).